

COMP0222 Coursework 02: Visual and LiDAR SLAM

Group 6

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27 April 2026

Camera Part: Public Visual SLAM

Placeholder for teammates

Add KITTI 07 and TUM monocular ORB-SLAM2 results here, including the effect of feature count, outlier rejection, and loop closure.

- Keep this slide to roughly 1 minute.
- Use report-submitted plots and metrics only.
- End with the main reason tracking improves or degrades.

Camera Part: Our Own Sequences

Placeholder for teammates

Add the indoor and outdoor RGB sequence collection procedure, COLMAP reconstruction, ORB-SLAM2 tracking, and EVO comparison.

Indoor

Add one representative frame, trajectory plot, and one sentence about the main difficulty.

Outdoor

Add one representative frame, trajectory plot, and one sentence about why tracking was more stable.

Camera Part: Results and Discussion

Placeholder for teammates

Add the final camera-side conclusion and transition into the LiDAR section.

- Mention that the camera and LiDAR sections use different sensors and different evidence.
- Leave enough time: the LiDAR part starts after roughly 5 minutes.

LiDAR Data Collection

Sequence	Scans	Time (s)
Large classroom	1234	96.90
Small room	659	60.64
Outdoor	4796	407.18

- RP-Lidar A2M12, carried roughly level
- Two loops, then return to the start
- Markers used only for checking

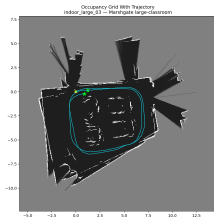
Main data challenge

The outdoor run is much longer and more open, so scan matching has less nearby structure and more accumulated drift.

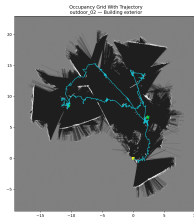
What we evaluate

Map shape, trajectory consistency, loop detection, and closure error.

Baseline Mapping and Parameter Study



Large classroom: stable walls and furniture returns.



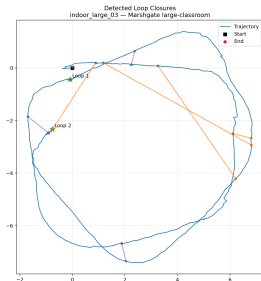
Outdoor: long route, visible drift.

Baseline: match each scan against the growing map, then build point-cloud and occupancy-grid maps.

Case	Class.	Room	Outdoor
Drift (m)	0.886	3.586	7.000

Parameter result: no single setting is best everywhere. Useful scan detail depends on scene scale and structure.

Loop-Closure Detection



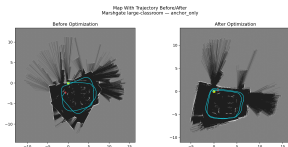
Large classroom loop detections.

Detection logic: compare keyframes against earlier places, reject nearby-in-time candidates, verify alignment, then deduplicate repeated matches.

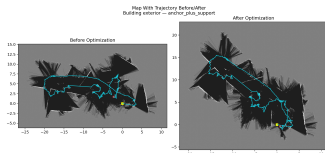
Sequence	Loops	Marker hits
Large classroom	7	2/2
Small room	4	1/2
Outdoor	12	0/2

Outdoor marker hits are weak, but repeated places are still found.

Pose-Graph Optimisation



Large classroom: final pose moves closer to the start.



Outdoor: two loop edges close the route more smoothly.

What changed

Odometry links keep the local path shape, and loop links pull revisited places back together.

Sequence	Before	After
Large classroom	2.443	0.898
Small room	2.395	0.546
Outdoor	2.820	0.817

Closure error in metres. Lower is better.

LiDAR Conclusion

Main result

Laser odometry can build useful maps, but loop closure is needed to make the second loop consistent.

- Parameter choices changed by environment.
- Loop detection found useful revisits in all three sequences.
- Optimisation reduced closure error for every sequence.

Sequence	Loop edges	Improvement
Large classroom	1	1.546 m
Small room	1	1.848 m
Outdoor	2	2.003 m

The outdoor result needs one supporting loop edge to close the route without bending it too much.

Thank You

Thank you

Questions?

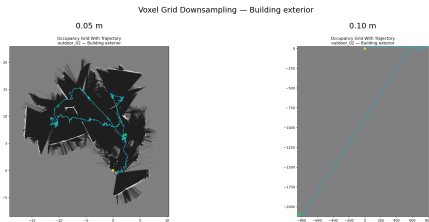
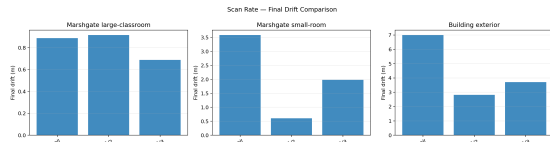
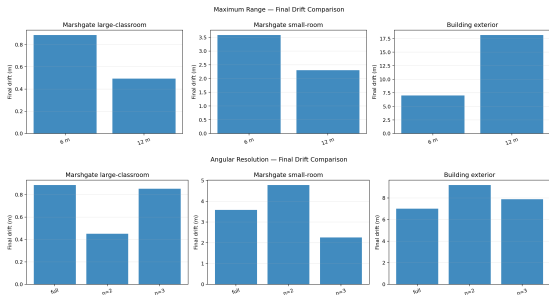
Group 6: David Zhong, Harry Tan, Yukun Wang

Backup: Parameter Study Details

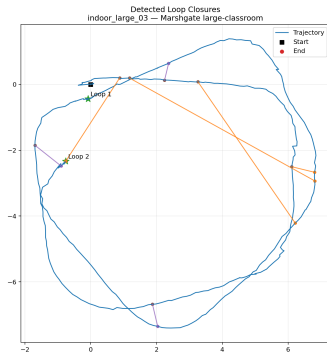
Study	Setting	Classroom	Room	Outdoor
Range	6 m	0.886	3.586	7.000
Range	12 m	0.494	2.301	18.183
Beams	Full	0.886	3.586	7.000
Beams	1/2	0.451	4.782	9.203
Beams	1/3	0.852	2.258	7.875
Voxel	0.05 m	0.886	3.586	7.000
Voxel	0.10 m	0.886	3.674	2251.157
Scans	All	0.886	3.586	7.000
Scans	1/2	0.915	0.608	2.820
Scans	1/3	0.687	1.978	3.708

Final drift in metres. The 0.10 m voxel outdoor case failed badly because too much local structure was removed.

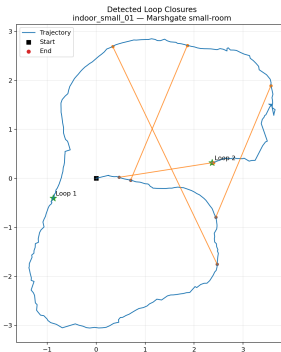
Backup: Parameter Figures



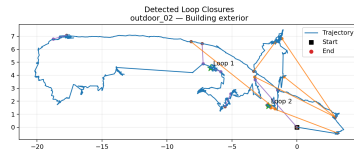
Backup: Loop Detections by Sequence



Large classroom: 7 loops, 2/2
marker hits

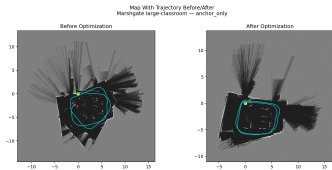


Small room: 4 loops, 1/2 marker
hits

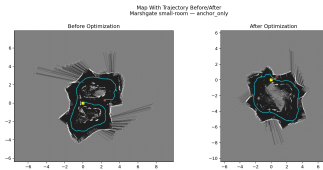


Outdoor: 12 loops, 0/2 marker hits

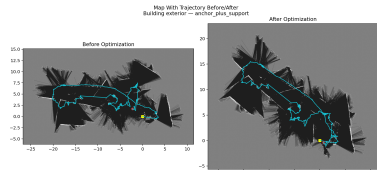
Backup: Optimisation by Sequence



Large classroom: 2.443 m to 0.898 m

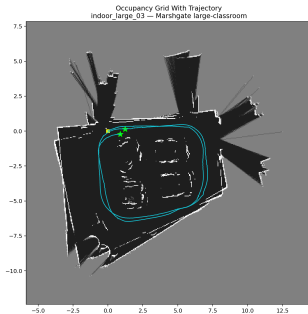


Small room: 2.395 m to 0.546 m

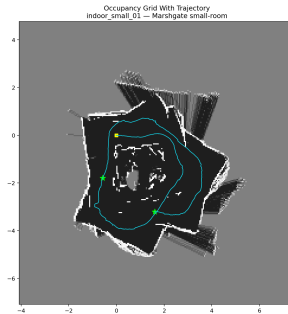


Outdoor: 2.820 m to 0.817 m

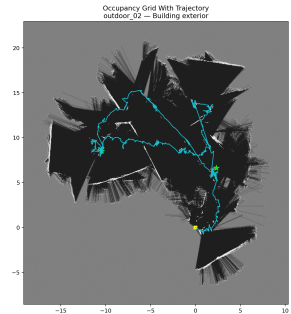
Backup: Baseline Maps by Sequence



Large classroom



Small room



Outdoor